

Nikhil Kudache

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EDUCATION

University of Southern California

Master of Science in Computer Science

- Relevant Coursework: Analysis of Algorithms, Database Systems, Machine Learning for Data Science, Applied Natural Language Processing

Los Angeles, CA

Aug 2025 – May 2027

Walchand Institute of Technology

Bachelor of Technology in Computer Science (GPA: 3.94/4.0)

- Relevant Coursework: Data Structures & Algorithms, OS, DBMS, OOP (Java/C++), Software Engineering, Cloud Computing, AI, Big Data.

Solapur, India

Aug 2018 – Jul 2022

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, SQL, Bash

Backend & APIs: REST APIs, FastAPI, Node.js, Next.js, JWT Authentication, Object-Oriented Design

Databases: PostgreSQL, MySQL, MongoDB, Azure Table Storage (NoSQL), Pinecone, Object Storage

Cloud & Infrastructure: AWS (EC2, S3, ECS, Lambda, IAM, VPC), Azure, GCP, Docker, Kubernetes, Terraform (IaC)

DevOps & CI/CD: Jenkins, GitHub Actions, CI/CD Pipelines

Tools & AI: Git, Linux/Unix, Postman, LangChain, RAG Pipelines, Multi-Agent Systems

EXPERIENCE

University of Southern California – Human Technology Interaction Lab

Software Engineer

Los Angeles, CA

Oct 2025 – Present

- Built a secure data ingestion platform for a clinical digital twin assistant, supporting 50+ surgeons, 600+ assigned image-review tasks, and 1,000+ stored image artifacts.
- Implemented the core review workflow for assigning images to surgeons, loading blob-backed image metadata, rendering category dependent question flows, and persisting structured responses.
- Integrated Azure Blob Storage with Managed Identity, RBAC-based access controls, GitHub Actions, and Azure App Service, securing 1,000+ artifacts with 0 public Blob Storage endpoints while enabling automated production releases.

Deloitte USI

Security Engineer - Cloud Infrastructure

Hyderabad, India

Jan 2023 – Jul 2025

- Built Python automation integrating Prisma Cloud and Jira APIs to close stale security tickets, clearing 300+ tickets and saving 20 hours/week of analyst effort.
- Developed reusable Terraform modules for AWS networking, IAM, and compute, standardizing cloud infrastructure deployment and reducing manual configuration errors across client teams.
- Integrated Checkov and Prisma Cloud scans into Jenkins CI/CD pipelines, blocking 90+ high-severity IaC and security violations before deployment.
- Troubleshoot production network and WAF policies on Palo Alto Panorama and Imperva, debugging live traffic issues to maintain availability for business-critical applications.

TECHNICAL PROJECTS

KiroGate | Python, FastAPI, Next.js, PostgreSQL, Docker, Kubernetes, AWS STS

June 2026 – Present

- Built KiroGate, a runtime policy gateway that moves AI-agent safety from prompt guardrails to deterministic tool-call enforcement before database, HTTP, or cloud actions execute.
- Implemented a 10-check enforcement pipeline for auth, schema validation, policy rules, egress control, quotas, scoped AWS STS credentials, response filtering, and audit logging.
- Validated the system with 4 attack scenarios and 218 pytest checks, blocking unsafe deletes, SSRF attempts, data exfiltration, and unauthorized tool calls with correlation-ID audit logs.

CourseMate RAG | Python, FastAPI, LangChain, Pinecone, Docker

June 2026

- Built CourseMate, a source-grounded RAG study assistant that ingests course PDFs, retrieves relevant passages through vector search, and returns LLM answers with page-level citations.
- Designed a provider-agnostic RAG architecture supporting 4 LLM providers, 3 embedding providers, and 2 vector stores, enabling model, embedding, and storage changes through configuration instead of code rewrites.
- Built an evaluation harness for lookup, multi-chunk synthesis, citation validation, and out-of-scope refusal, testing whether answers stay grounded in uploaded course material.

AWARDS & RECOGNITION

Best Use of Miro x Kiro – Miro x Kiro LA Hackathon

Jun 2026

- Won Best Use of Miro x Kiro for building Dietary Maps AI, a natural-language restaurant search engine that verifies dietary claims across multiple sources and returns results with transparent, evidence-backed confidence scores.
- Architected a multi-agent AI pipeline in Next.js 14 + TypeScript with a weighted confidence model (evidence quality, review consistency, menu verification, data recency), shipped as a deployed Vercel app.

Applause Award – Deloitte

Apr 2025

- Recognized for building the automation system that eliminated 30% of manual task assignments and saved the cyber infrastructure team 20 hours per week.